



$$Q_s = -30 + 6(\underline{P-5})$$

$$Q_d = -60 + 6P$$

$$f(x) = \sqrt{x} + \frac{1}{x}$$

$$f'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x^2}$$

$$f(x) = \frac{x \cdot \left(\frac{1}{2\sqrt{x}} - \frac{1}{x^2} \right)}{\sqrt{x} + \frac{1}{x}} = \frac{x \left(\frac{x^2 - 2\sqrt{x}}{2\sqrt{x} \cdot x^2} \right)}{\frac{x\sqrt{x} + 1}{x}}$$

$$\frac{\cancel{x}(x^2 - 2\sqrt{x})}{2\sqrt{x} \cdot \cancel{x^2}} \cdot \frac{\cancel{x}}{(x\sqrt{x} + 1)} =$$

$$\frac{x^2 - 2\sqrt{x}}{2\sqrt{x}(x\sqrt{x} + 1)} = \frac{x^2 - 2\sqrt{x}}{2x \cdot x + 2\sqrt{x}} =$$

$$\frac{x^2 - 2\sqrt{x}}{2x^2 + 2\sqrt{x}} = \frac{4 - 2\sqrt{2}}{8 + 2\sqrt{2}}$$

$$y = x^2 \ln x$$

$$y' = 2x \ln x + \cancel{\frac{1}{x}} \cdot x^2 = 2x \ln x + x$$

$$P(x) = \frac{\cancel{x} \cdot (2x \ln x + x)}{x^2 \ln x} = \frac{\cancel{x} (2 \ln x + 1)}{x \ln x}$$

$$\frac{2 \ln x + 1}{\ln x} = 0$$

$$2 \ln x + 1 = 0$$

$$2 \ln x = -1$$

$$\ln x = -\frac{1}{2}$$

$$\log_e x = -\frac{1}{2}$$
$$x = e^{-\frac{1}{2}}$$

$$\frac{2\ln x + 1}{\ln x} > 1$$

$$\frac{2\ln x + 1}{\ln x} - 1 > 0$$

$$\frac{2\ln x + 1 - \ln x}{\ln x} > 0$$

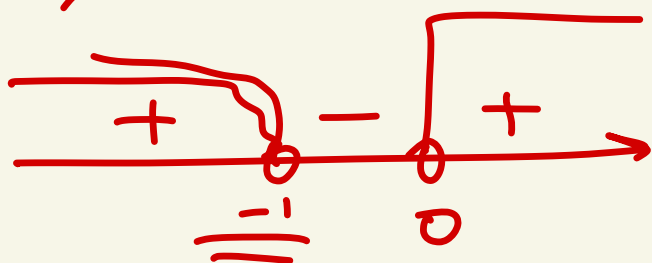
$0 < 3 \log$

$$\frac{\ln x + 1}{\ln x} > 0$$

$x > 0$

$$\ln x = t$$

$$\frac{t+1}{t} > 0$$



$$\ln x > 0$$

$$x > e^0 = 1$$

$$x \in (0; \frac{1}{e}) \cup$$

$(1; +\infty)$

$$\ln x < -1$$

$$x < e^{-1} = \frac{1}{e}$$

$$Q_d = 200 - 5P$$

2) ~~ноoooo~~ ~~затра~~ $P = 20$?

2) P ? $E > 1$

$$Q_d = -5$$

$$\frac{-5P}{200 - 5P} > 1$$

$$\frac{-5P}{200 - 5P} - 1 > 0$$

$$\frac{-5P - (200 - 5P)}{200 - 5P} > 0$$

$$\frac{\cancel{-5P} - 200 + \cancel{5P}}{200 - 5P} > 0$$

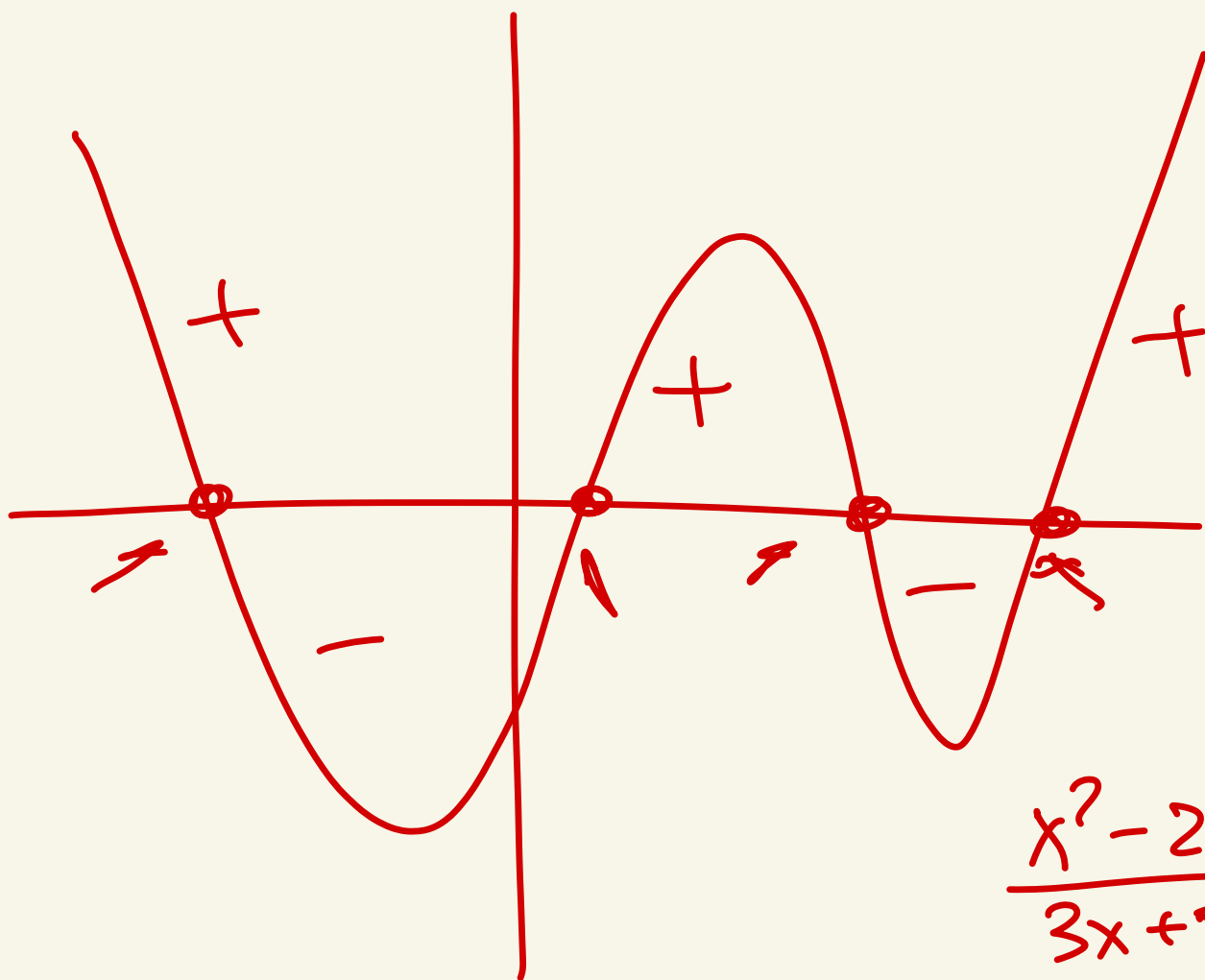
$$\frac{-200}{200 - 5P} > 0$$

$$\frac{200}{5} = 40$$

$$P \geq 40$$

$$\frac{-5P}{200-5P} = 1$$

$$\frac{-5P + 200 + 5P}{200 - 5P} = 0$$



$$\frac{x^2 - 2x}{3x + 5}$$

$$\text{Выручка} = P \cdot Q_d =$$

$$P = 20$$

$$(P \cdot (200 - 5P))$$

$$P = 10$$

$$20 \cdot (200 - 100) =$$

$$P = 0$$

$$20 \cdot 100 = \underline{\underline{2000}}$$

$$P = 30$$

$$10 \cdot (150) = 1500$$

$$30 \cdot (50) = 1500$$

$$f'(P) = (200P - 5P^2)' = 200 - 10P$$

$$f'(P) = 0 \rightarrow \text{max выруч}$$

$$200 - 10P = 0$$

$$P = \frac{200}{10} = 20$$

$$f'(x) > 0$$

$$\Rightarrow f(x) \text{ возр}$$

$$f'(x) < 0$$

$$\Rightarrow f(x) \text{ убав}$$

$$f'(x) = 0$$

$$\Rightarrow \text{экстремум!}$$

max
min

$$\underline{\underline{Q_d = 100 - 2P}}$$

$$|E| = \left| \frac{P}{Q} \right| = \frac{5}{5} = 1$$

$$1) \text{ Энелер} = 0$$

$$2) \text{ Энелер} > 1$$

$$3) \text{ макс вырлзу?}$$

$$100 - 4P = 0$$

$$P = 25$$

$$\frac{-2P}{100 - 2P} = 0$$

$$(P = 0) \quad 100 \neq 2P \quad P \neq 50$$

$$\frac{1\% X}{9\% Y} = E = 0 = 5$$

$$= 1$$

$$Y = 2X + 100 \quad Y' = 2$$

$$\frac{X \cdot 2}{X} = 2$$

$$y = x$$

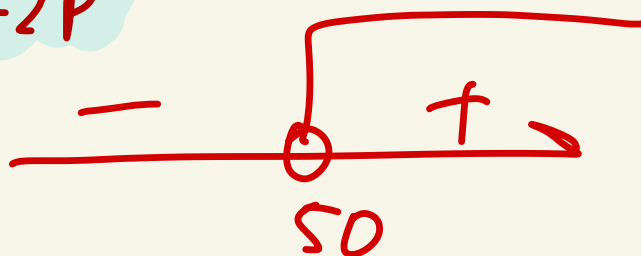
$$y = x + 100$$

$$y = 2x$$

y'	значения
1	1
1	$\frac{x}{x+100}$
2	$\frac{2x}{2x} = 1$

$$\frac{-2P}{100-2P} > 1$$

$$\frac{-100}{100-2P} > 0$$



$$\frac{-100}{+100}$$

$$Q_2 = 50 - 10P$$

$$P? \quad E = -4$$

$$E = \frac{-10P}{50-10P}$$

$$\frac{-10P}{50-10P} = -4$$

$$\frac{-10P + 4}{50-10P}$$

$$\frac{-10P + 4(50-10P)}{50-10P} = 0$$

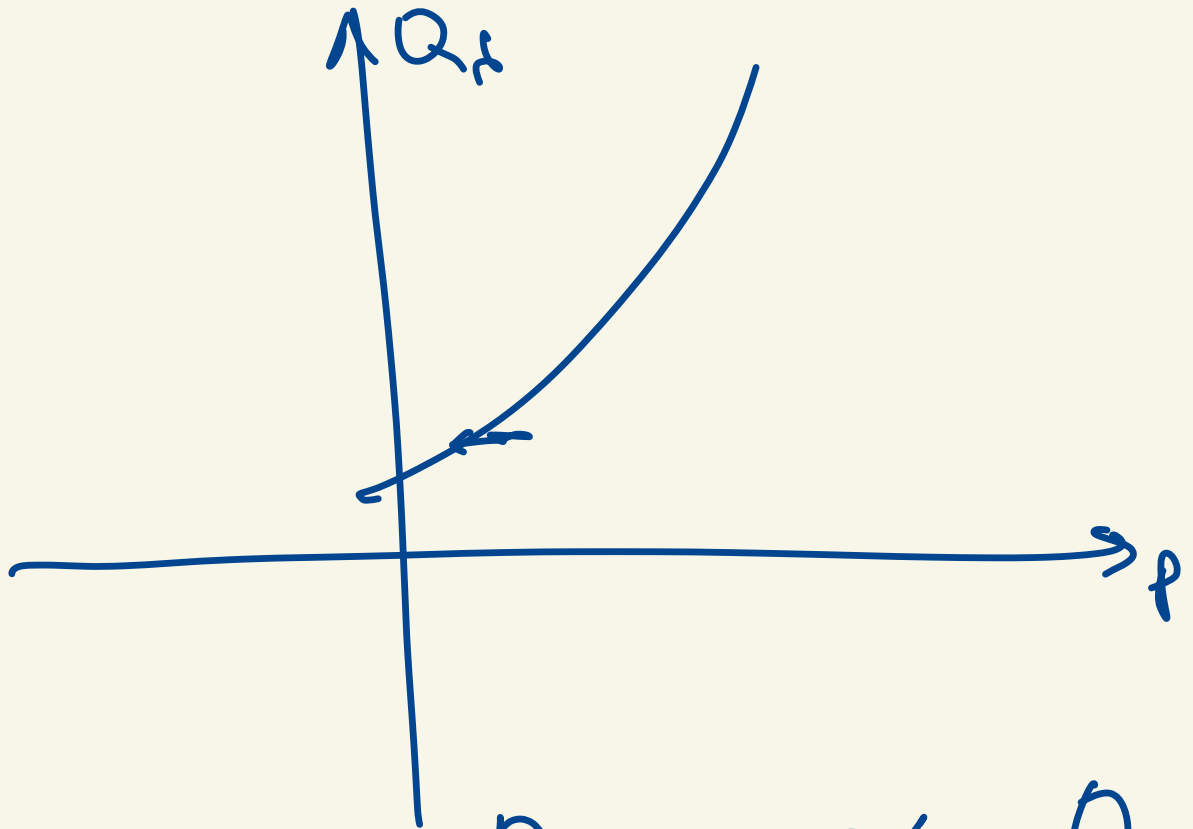
$$\frac{-10P + 200 - 40P}{50-10P} = 0$$

$$\frac{-50P + 200}{50-10P} = 0$$

$$E_d = \frac{\Delta Q \%}{\Delta P \%}$$

$$P \rightarrow 2\% \rightarrow Q \rightarrow 5\%$$

$$E_d = \frac{5\%}{2\%} = \frac{5}{2} = 2,5$$



$$\Delta Q\% = Q_{\text{ср}}\% - Q_{\text{бн}}\%$$

1100
1000
110%
100%

$$E_d = -2$$

$p \uparrow$ на 3%

$Q_s?$

$$\frac{\Delta Q\%}{\Delta p\%} = -2$$

$$\Delta Q \% = \Delta P \cdot (-2)$$

$$T = P \cdot Q = -6\%$$

изменение выручки

$$\Delta T = \Delta P \% + \Delta Q \% =$$

$$3\% - 6\% = -3\%$$

$$T = 0,4 K^{1,06}$$

расх. на строит

ОТВЕТИ
График

1. Найти эластичность ?

$$T'(K) = 0,4 \cdot 1,06 K^{0,06}$$

$$E(K) = \frac{0,4 \cdot 1,06 \cdot K^{0,06} \cdot K^1}{0,4 \cdot K^{1,06}} =$$

$$= \frac{0,4 \cdot 1,06 \cdot \cancel{k^{1,06}}}{0,4 \cdot \cancel{k^{1,06}}} = 1,06$$

расх $\downarrow$ 1%
T% ?

$$E = \frac{\Delta Q \%}{\Delta P \%}$$

Q(P)
J(K)

$$E = \frac{\Delta T \%}{\Delta K \%}$$

Было 98%.
Стало 97%.

$$97\% - 98\% = -1\%$$

$$E = 1,06 = \frac{\Delta T \%}{-1}$$

$$\Delta T \% = -1 \cdot 1,06 = -1,06\%$$

расх. увелич на 5%.

Tz.